

Lab Project 01: Library Database Schema Design

Target Audience: Computer Science & Engineering Students

Environment: XAMPP, phpMyAdmin, MySQL

1. Project Objective

This structural laboratory project outlines the end-to-end framework required to establish a local database management environment and program relational structures. By completing this assignment, students will develop core competencies in:

- Configuring and operating a standalone server infrastructure leveraging the XAMPP architecture ecosystem.
- Authoring and deploying standard SQL Data Definition Language (DDL) declarations.
- Enforcing referential schema constraints using programmatic Primary Key and Foreign Key operations.
- Verifying systematic structural alignment visually inside the internal storage designer platform.

2. Environment Setup & Prerequisites

Prior to formulating code statements, the local developer configuration must be executed with precision according to the procedures detailed below.

2.1 Installing the Environment

1. Acquire the official installation binary for XAMPP (Windows architecture variation) via the authenticated distribution site.
2. Initialize the downloaded executable asset. Acknowledge and clear any access privilege warnings presented by User Account Control.
3. Advance through the setup utility, ensuring the absolute file storage configuration is preserved at its default target path: `C:/xampp`. Conclude execution.

2.2 Activating Relational Engine Subsystems

1. Launch the XAMPP Control Panel component interface.
2. Locate the explicit modular listings for both the Apache server subsystem and the MySQL relational gateway engine. Trigger the initialization process by clicking the respective Start inputs.
3. Verify operational stability by inspecting the active state markers. The module descriptions will reflect a green indicator background, confirming active background assignment allocations.

3. Database Implementation (SQL Script)

Initialize an authenticated system browser session and map navigation targets to `localhost/phpmyadmin` (alternatively, utilize the Admin entry vector directly within the operational engine panel). Relocate interaction coordinates onto the central SQL processing workspace to deploy the subsequent procedural query blocks sequentially.

3.1 Space Allocation and Initialization

Construct an isolated relational schema partition optimized specifically for the library information workflow environment.

```
CREATE DATABASE LibraryDB;  
USE LibraryDB;
```

Execute the control script block by clicking the Go action button positioned inside the console platform interface.

3.2 Declaring Core Structural Entities

Establish the underlying primary tables responsible for retaining persistence attributes regarding inventory details (Books) and cohort directories (Students). Implement distinct transactional identification fields utilizing explicit `PRIMARY KEY` integrity boundaries.

```
-- Define Core Inventory Entity  
CREATE TABLE Books (  
    book_id INT NOT NULL,  
    title VARCHAR(100),  
    author VARCHAR(100),  
    PRIMARY KEY (book_id)  
);  
  
-- Define Core Cohort Entity  
CREATE TABLE Students (  
    student_id INT NOT NULL,  
    student_name VARCHAR(100),  
    PRIMARY KEY (student_id)  
);
```

3.3 Constructing Relational Bridges

Implement the dependent intersection schema construct named `Borrow`, mapped explicitly to log transactional interactions recording asset checking events. This architecture secures entity logic parity by mapping constrained references back onto the parent tables.

```
-- Define Relational Transaction Intersect
CREATE TABLE Borrow (
    borrow_id INT NOT NULL,
    book_id INT,
    student_id INT,
    borrow_date DATE,
    PRIMARY KEY (borrow_id),
    FOREIGN KEY (book_id) REFERENCES Books (book_id),
    FOREIGN KEY (student_id) REFERENCES Students (student_id)
);
```

System Note on Data Integrity: The implementation of the `FOREIGN KEY` constraint systematically prevents the processing of corrupted tracking data. The architecture engine will actively abort transactions inside the tracking array if matching identification elements are missing from the master structures.

4. Visual Schema Verification

Execute structural diagnostics to validate relational integrity mappings across all generated table nodes:

1. Access the global object storage hierarchy tree in the left navigation section and activate the specific `LibraryDB` schema context.
2. Expose extended operations by selecting the More command element positioned across the secondary option cluster.
3. Select the Designer interactive canvas variant from the context list.
4. The system will output a clean entity-relationship canvas outlining the interactive states for the `Books`, `Students`, and `Borrow` properties, structurally bounded via clean relational vector channels.

5. Assignment Submission Requirements

To secure evaluation metrics, candidates must structure an explicit file portfolio containing the specified artifacts below:

- **Source Manifest:** An unmodified text code file containing all structured declarative queries stored explicitly with a `.sql` or `.txt` file indicator extension.
- **Operational Validation:** A system viewport capture demonstrating the running environment state, validating active parameters for both structural engine components.
- **Architecture Mapping:** A fullscreen render showing the structural relational nodes within the operational Designer perspective context.
- **Packaging:** Group the combined artifacts inside a zip database container formatted as `StudentID_DBMS_Project01.zip` and deliver it via the institutional course management portal.

6. Troubleshooting Guidelines (FAQ)

Question: The engine modules systematically report failure metrics or drop operational connections immediately upon initiation attempts?

Resolution: This behavior typically points to networking socket collisions, where secondary services occupy port channels 80 or 3306. Terminate conflicting service processes, execute a clean workstation refresh, and restart the environment console.

Question: The execution console yields a structural validation failure error during the generation of the relational mapping block?

Resolution: Review data profile declarations. The targeting data parameters used inside the child entity table must correlate exactly with the primitive attributes defined on the master key entities. Any divergence in the data profile will halt schema execution.